

Scenario-Based Fully Dressed Use Cases

Scenario 1: Buy Stocks Over the Web

Use Case: Buy Stocks Over the Web

Primary Actor: Purchaser

Scope: Finance Package (PAF)

Level: User goal

Stakeholders and Interests:

- Purchaser: wants to buy stocks and update portfolio.
- Stock Agency: wants full purchase info.

Precondition: User is logged in.

Minimal Guarantee: Logs sufficient information for error detection.

Success Guarantee: Purchase is acknowledged; portfolio and logs updated.

Main Success Scenario:

1. Purchaser selects to buy stocks.
2. System obtains website choice from user.
3. System opens web connection.
4. Purchaser browses and buys stock.
5. System intercepts responses and updates portfolio.
6. System displays new portfolio standing.

Extensions:

- 2a: Website unsupported → user suggests alternative or cancels.
- 4a: Purchase delayed → system logs delay and prompts user later.

Special Requirements:

- Touchscreen UI visible from 1 meter.
- Credit authorization within 30 seconds (90% cases).
- Robust recovery if remote services fail.

Scenario 2: Process Sale at POS

Use Case: Process Sale

Primary Actor: Cashier

Scope: Retail POS System

Level: User goal

Stakeholders and Interests:

- Cashier: wants to process sales quickly.
- Store: wants accurate transaction logging and inventory updates.

Precondition: Cashier is logged in; POS is ready.

Minimal Guarantee: Sale information is logged even if payment fails.

Success Guarantee: Sale is completed; inventory and accounting updated.

Main Success Scenario:

1. Cashier scans items.
2. System calculates total, taxes, and discounts.
3. Customer pays using cash, card, or digital wallet.
4. System validates payment.
5. System prints receipt and updates inventory.

Extensions:

- 3a: Payment declined → system prompts for alternative payment.
- 1a: Item barcode not found → system allows manual entry.

Special Requirements:

- Receipt printer must be functional.
- System response time ≤ 2 seconds per item.
- Multilingual support for UI.

Scenario 3: Borrow a Book (Library Management System)

Use Case: Borrow a Book

Primary Actor: Librarian

Secondary Actor: Library Member, Library Management System

Scope: Library Management System

Level: User goal

Preconditions:

- Librarian is logged into the system.

- Member is registered in the system and has an active account.
- Book exists in the library catalog.

Basic Flow (Main Success Scenario):

1. Member requests to borrow a book.
2. Librarian searches for the book in the catalog.
3. System checks the availability of the book.
4. System checks the member's account for overdue books or fines.
5. If the book is available and the member has no issues, the system records the borrowing transaction.
6. System updates the book's status as borrowed.
7. System prints or sends a borrowing receipt to the member.
8. Member receives the book.

Alternative Flows / Exceptions:

- 3a: Book is not available → system notifies librarian; may reserve book for member.
- 4a: Member has overdue books or unpaid fines → borrowing blocked until resolved.
- 5a: System error during transaction → system logs error; librarian retries or escalates.

Postconditions / Success Guarantee:

- Book status updated as borrowed.
- Member account reflects borrowed book and due date.
- Transaction recorded for reporting.

Scenario 4: Return a Book (Library Management System)

Use Case: Return a Book

Primary Actor: Librarian

Secondary Actor: Library Member, Library Management System

Scope: Library Management System

Level: User goal

Preconditions:

- Librarian is logged in.

- Member has borrowed the book.

Basic Flow (Main Success Scenario):

1. Member presents the book to the librarian.
2. Librarian scans or enters book details into the system.
3. System verifies the book is borrowed by the member.
4. System updates the book status as returned.
5. System calculates any late fees.
6. System prints or sends a receipt to the member.

Alternative Flows / Exceptions:

- 3a: Book is not recognized → system prompts manual verification.
- 5a: Member has outstanding fines → system alerts librarian; payment processed or borrowing blocked.
- 4a: System error → transaction logged; librarian retries or escalates.

Postconditions / Success Guarantee:

- Book is marked as available in the catalog.
- Member account updated with return date and fines (if any).
- Transaction recorded for reporting.

Scenario 5: Register a New Member (Library Management System)

Use Case: Register a New Member

Primary Actor: Librarian

Scope: Library Management System

Level: User goal

Preconditions:

- Librarian is logged in.
- Prospective member meets library membership requirements.

Basic Flow (Main Success Scenario):

1. Librarian selects “Register New Member” in the system.
2. Librarian enters member personal details (name, address, contact).
3. System validates inputs and checks for existing records.

4. System creates member account and assigns member ID.
5. System prints membership card or sends confirmation to member.

Alternative Flows / Exceptions:

- 3a: Member already exists → system notifies librarian; registration aborted.
- 3b: Invalid input → system requests correction.
- 4a: System error → transaction logged; librarian retries or escalates.

Postconditions / Success Guarantee:

- Member account successfully created.
- Member receives ID and confirmation.
- System logs registration for auditing.

Scenario 6: Reserve a Book (Library Management System)

Use Case: Reserve a Book

Primary Actor: Librarian

Secondary Actor: Library Member, Library Management System

Scope: Library Management System

Level: User goal

Preconditions:

- Librarian is logged in.
- Member is registered and has an active account.
- Book exists in catalog.

Basic Flow (Main Success Scenario):

1. Member requests to reserve a book.
2. Librarian searches for the book in the catalog.
3. System checks the availability of the book.
4. System records reservation details (member, book, date).
5. System confirms reservation to the member.

Alternative Flows / Exceptions:

- 3a: Book is available now → system suggests immediate borrowing instead of reservation.
- 4a: Member has unpaid fines → reservation blocked until fines are cleared.

- 4b: System error → logs error; librarian retries or escalates.

Postconditions / Success Guarantee:

- Reservation recorded successfully in system.
- Member notified of reservation confirmation.
- Book will be held until pickup or reservation expires.

Scenario 7: Online Food Order (Restaurant Management System)

Use Case: Place Online Food Order

Primary Actor: Customer

Secondary Actor: Online Ordering System, Delivery Staff

Scope: Restaurant Management System

Level: User goal

Preconditions:

- Customer is logged into the restaurant app.
- Menu items are updated and available.

Main Success Scenario:

1. Customer browses menu and selects items.
2. System calculates subtotal, taxes, and delivery charge.
3. Customer provides delivery address and payment method.
4. System processes payment and confirms order.
5. Restaurant receives order and prepares food.
6. Delivery staff picks up and delivers the order.
7. System marks order as completed and sends receipt.

Alternative Flows / Exceptions:

- 3a: Invalid address → system prompts correction.
- 4a: Payment declined → system prompts alternate method.
- 5a: Item unavailable → system offers replacement or refund.
- 6a: Delivery delayed → system notifies customer and updates ETA.

Postconditions / Success Guarantee:

- Order status updated as completed.

- Payment recorded; inventory adjusted.
- Customer receives receipt and feedback prompt.

Special Requirements:

- Order confirmation within 10 seconds.
- Payment gateway response time ≤ 5 seconds.
- Real-time delivery tracking.

Scenario 8: Book Flight Ticket (Airline Reservation System)

Use Case: Book Flight Ticket

Primary Actor: Passenger

Secondary Actor: Airline Reservation System, Payment Gateway

Scope: Airline Reservation System

Level: User goal

Preconditions:

- Passenger has an active account.
- Flights and seat availability are updated in the system.

Main Success Scenario:

1. Passenger searches for available flights by date and destination.
2. System displays available flights and fare options.
3. Passenger selects a flight and enters passenger details.
4. Passenger proceeds to payment.
5. System processes payment and confirms booking.
6. System sends electronic ticket (e-ticket) via email/SMS.

Alternative Flows / Exceptions:

- 2a: No flights available → system suggests nearby dates or airports.
- 4a: Payment declined → system prompts for another payment option.
- 5a: Seat sold out during payment → system refunds and suggests alternatives.

Postconditions / Success Guarantee:

- Ticket booked and stored in passenger's account.
- Payment and booking details recorded in system.

Special Requirements:

- Secure payment (SSL encryption).
- Booking confirmation within 10 seconds.
- System handles up to 500 concurrent bookings.

Scenario 9: Submit Online Assignment (E-Learning System)

Use Case: Submit Online Assignment

Primary Actor: Student

Secondary Actor: Instructor, LMS (Learning Management System)

Scope: E-Learning Platform

Level: User goal

Preconditions:

- Student is logged in to LMS.
- Assignment is published and open for submission.

Main Success Scenario:

1. Student opens the course and selects the assignment.
2. System displays instructions and due date.
3. Student uploads the completed file.
4. System verifies file format and uploads to storage.
5. System records submission timestamp and notifies instructor.

Alternative Flows / Exceptions:

- 3a: File format invalid → system prompts correction.
- 4a: Network error → system retries upload or saves draft.
- 5a: Submission after deadline → system flags late submission.

Postconditions / Success Guarantee:

- Assignment successfully stored.
- Submission time logged.
- Notification sent to instructor.

Special Requirements:

- File upload limit: 50 MB.
- System response within 3 seconds.
- Auto-save draft every 60 seconds.